

Machine Learning Prediction for Experimental Quench Voltage of HTS Coils using a combined Framework with Time-series Generative Adversarial Network and Long Short-Term Memory Techniques

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Abstract— Fast and reliable quench detection is critical for protecting high temperature superconducting coils, magnets, and devices, yet experimental datasets remain limited due to cryogenic constraints and risk of conductor damage. This limitation is one of the key challenges in developing machine learning (ML) techniques for predicting quench through regression-based forecasting of voltage signals. To address this, we propose a framework that combines Time-series Generative Adversarial Networks (TimeGAN) for artificial sequence generation with Long Short-Term Memory (LSTM) predictors. From 36 measured voltage sequences, TimeGAN generated 90 artificial sequences to expand the dataset and mitigate overfitting. Augmenting the LSTM improved quench prediction accuracy to ~3% mean relative error. This study demonstrates the first application of TimeGAN to superconducting quench signals and highlights the benefit of ML-driven augmentation for improving quench prediction robustness.

Index Terms— Artificial intelligence, LSTM, ML, Quench prediction, Quench voltage estimation, TimeGAN.

I. INTRODUCTION

High-temperature superconducting (HTS) technology is promising for being adopted in fusion industry, renewable energy, electric aerospace system, and power network devices, where HTS coils play a critical role in enabling compact geometries, high current and power densities [1], [2]. However, HTS coils are inherently vulnerable to quench events, a sudden transit from superconducting state to normal metal state that leads to localized heating and potentially catastrophic damage [3], [4]. Because the high-scale stored energy and non-linear dynamics involved, fast protection schemes are critical for superconducting devices including magnets: early and reliable quench detection and potentially protection allows superconducting components to be safe and prevents irreversible degradation of the coated conductor. Conventional voltage-threshold and bridge-balance methods are widely used but often suffer from noise sensitivity, inductive effects, and delayed detection in HTS coils with slow normal-zone propagation [3], [5], [6]. Among traditional

methods, the bridge-balance voltage method is most widely used [7], [8], [9], in which sections of the superconductor form a Wheatstone bridge to cancel inductive voltages during current ramping and isolate the small resistive signal that marks quench onset. Other approaches include Hall probes for detecting local magnetic-field variations and current-redistribution monitoring at multiple points [10], [11], [12]. Recently, artificial intelligence (AI) and machine learning (ML) techniques have been applied to different problems within superconductivity [13], [14], [15], including quench detection by learning directly from voltage signals [16], [17], [18], [19], [20]. These quench detection techniques are based on regression tasks of AI/ML techniques and are different from signal processing-based detection approaches [3], [21], [22]. Although these approaches show advantages, a key limitation remains: they offer limited interpretability and raising concerns about trustworthiness in safety-critical superconducting systems [23], [24].

To address these challenges, this paper offers a framework that explicitly models temporal dynamics to generate realistic artificial sequences and demonstrates that augmented training enhances the reliability of AI-based forecasting. The results highlight its potential to improve predictive performance when experimental datasets are limited.

This paper makes the following contributions:

- 1) Artificial data generation: We present the first application of the Time-series Generative Adversarial Network (TimeGAN) to superconducting systems. By learning the temporal statistics of measured voltage signals from HTS coils, TimeGAN generates realistic and diverse artificial sequences under specific operating conditions. These synthetic signals expand limited datasets and reduce risk of overfitting.
- 2) Improved quench prediction: We show that combining artificial and real data in a Long Short-Term Memory (LSTM) predictor enhances early quench warning capability. Models trained on both data types achieve greater robustness and accuracy than those trained on real data alone.

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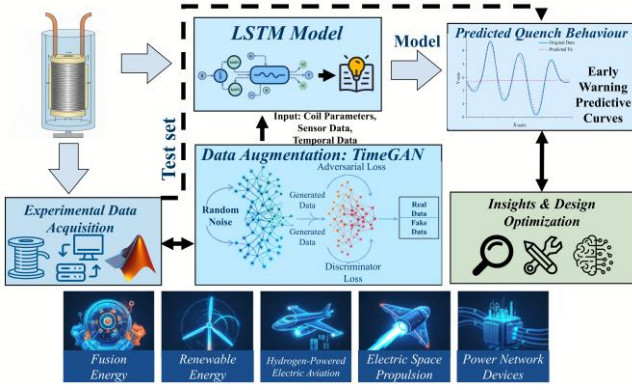


Fig.1. Schematic workflow of AI/ML-assisted quench prediction proposed in this paper.

II. HTS COIL SPECIFICATION & EXPERIMENTAL DATASET

The experimental dataset used in this study was obtained from an HTS coil [25], designed to reproduce the quench events. The coil was wound on a G10 former (90 mm diameter) and immersed in a liquid nitrogen (LN₂) bath at 77 K. It consisted of 13 turns of 12-mm-wide tape (with 1 μ m superconducting layer and Ni-5at%W, 75 μ m substrate and stainless-steel stabilizer), with total length of $\sim$ 3.5 m. The HTS tape and the coil exhibit a critical current (I_c) of 241 A and 251.7 A at 77 K, respectively.

To trigger quench events, the transport AC current exceeded the coil's critical limit. Test currents ranged from $\sim$ 250 A up to 490 A, producing both normal and quench regimes. Before executing an abrupt 100-ms quench-inducing high-current application, each test began with a stable 100-ms normal superconducting phase. Voltage taps captured the growth of local resistive zones, while a Rogowski coil measured the transport current. All signals were digitized by a high-speed data acquisition system, providing synchronized voltage and current waveforms.

The dataset comprises 36 temporal sequences, each representing a 0.1 s window of coil behaviour. Three channels were recorded: two voltages from different layer locations and the current. Each sequence contains 200 time-steps (sampling rate 2 kHz), which is sufficient to capture the transition from superconducting to resistive state.

III. QUENCH PREDICTION METHODOLOGY

Quench datasets are limited due to the high cost, cryogenic limitations, and risk of HTS tape damage in real coil tests. This data scarcity restricts training of data-driven models, making artificial quench data generation (data augmentation) a practical solution. However, conventional augmentation fails to capture the temporal dynamics critical for distinguishing quench from normal states. Variational Autoencoders often yield overly smooth signals [26], [27], while classical GANs cannot maintain sequential dependencies [28]. To address these limitations, we implemented TimeGAN, which is specifically designed for synthesizing realistic sequential data for quench cases. **Fig. 1** depicts the overall workflow of quench voltage prediction in this paper.

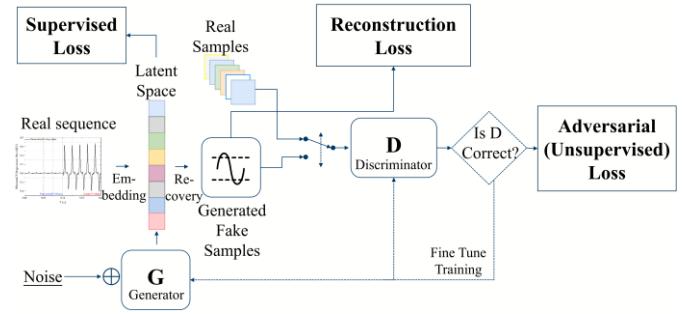


Fig.2. Architecture of the TimeGAN framework.

A. TimeGAN

TimeGAN provide a framework for generating artificial sequential data that preserves both temporal dynamics and statistical distributions of the original HTS coil quench dataset [28]. By integrating a recurrent structure with a supervised objective, TimeGAN captures temporal correlations, making it suitable for superconducting signals that show both short-term fluctuations and long-term transitions from normal to quench states.

As illustrated in **Fig. 2**, TimeGAN merges the autoencoder and GAN architectures [29]. It contains embedder and recovery process, generator, supervisor, and discriminator. Embedder and recovery process map original time-series X into a lower-dimensional latent representation H , and reconstruct it back to $\hat{X}$. Generator and supervisor generate synthetic latent sequences from random noise Z . The generator learns to produce synthetic representation H . The supervisor provides an additional recurrent step, predicting future latent states given the past, thereby encouraging long-range consistency. The discriminator distinguishes between real and synthetic latent trends. TimeGAN training involves four different types of composite loss. Reconstruction loss $\mathcal{L}_E$ ensures the embedder–recovery pair can faithfully reconstruct the input sequence. Supervised loss $\mathcal{L}_S$ encourages the supervisor's predictions of future latent states to match true latent dynamics. Adversarial loss $\mathcal{L}_G$ and $\mathcal{L}_D$ constitute an adversarial game between generator–supervisor and discriminator, aligning artificial and real latent distributions. Finally, moment matching loss $\mathcal{L}_M$ aligns the first and second statistical moments (mean and variance) of artificial sequences with real data, which reduces distributional mismatch and stabilizes training [28].

The overall generator objective combines all components [28]:

$$\mathcal{L}_{gen} = \mathcal{L}_G + \lambda_S \mathcal{L}_S + \lambda_M \mathcal{L}_M, \quad (1)$$

with hyperparameters λ_S, λ_M controlling the balance between realism and diversity.

Training proceeds in three stages: 1) Embedding pre-training optimizes reconstruction loss only, to stabilize the latent space. 2) Supervised pre-training trains the supervisor on latent dynamics without adversarial feedback. 3) Joint training optimizes synchronous optimisation of all components by alternately updating the generator/supervisor and discriminator. The trained TimeGANs generated artificial sequences with the same dimensionality as the original signals. We evaluated the generated signals using two complementary metrics:

1. The discriminative score measures the ability of a post-hoc recurrent neural network (RNN) classifier to distinguish

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between real and artificial sequences, with lower values indicating greater similarity [30].

2. The predictive score assesses the performance of an RNN trained on artificial data and tested on real data, where a lower mean absolute error reflects better preservation of temporal dependencies [28].

B. LSTM

While TimeGAN is responsible for generating artificial time-series data, it is important to verify that the data generated preserves the temporal dependencies relevant for AI modelling. The LSTM predictor captures temporal correlations through gated recurrent units and is well suited for modelling the normal-to-quench transition [5], [17], [18]. Training used MSE loss with Adam and early stopping.

As illustrated in **Fig. 3**, the LSTM predictor is an RNN consisting of stacked LSTM layers, each of which maintains a hidden state and a memory cell. At each time step t , the network processes an input vector x_t (voltage channel) and updates its hidden state h_t via three gating operations: input, forget, and output gates. These gates regulate information flow, allowing the model to retain or discard past signals as appropriate [18], [30]. The final hidden representation is passed through a fully connected output layer to forecast the next-step signal.

Performance was then assessed on a real testing set to measure the transferability of artificial data. The training loss was mean squared error (MSE), optimized with Adam optimizer. Early stopping was applied to prevent overfitting. To quantify fidelity, we used MAE to evaluate the difference between the predicted and true real test values. A low predictive score indicates that the artificial data encode temporal dependencies similar to real data, making them effective for training predictive models.

C. Metrics

We train a binary classifier $D(x)$ to distinguish between real sequences X and artificial sequences $\tilde{X}$ from TimeGAN. $D(x)$ was trained only on real sequences from the training split and the synthetic sequences generated from that same split by TimeGAN. No test sequences were used, ensuring there is no information leakage.

The discriminative score (DS) is defined as the classification accuracy on a held-out test set [28]:

$$DS = \frac{1}{N} \sum_{i=1}^N 1(D(x_i) = y_i) - 0.5, \quad (2)$$

where N = number of samples, $y_i \in \{0,1\}$ is the true label (real vs. synthetic). A score close to 0 (random guessing) means the artificial sequence is indistinguishable from real. A higher score ($\gg 0$) means the discriminator easily separates them, which is poor realism [31].

We also use the predictive score (PS) and a supervised predictor f_θ to assess how well a model trained on synthetic data generalises to real sequences [28]:

$$PS = \frac{1}{N(T-1)} \sum_{i,t} |x_{t+1} - P(x_{1:t})|, \quad (3)$$

where $x_{1:t}$ is the past window of a real sequence, x_{t+1} is the true next data point. A lower predictive score indicates that models trained on artificial sequences generalize well to real temporal dynamics.

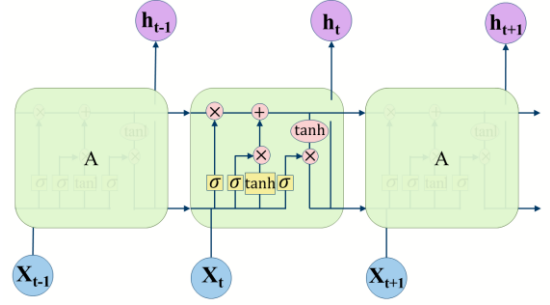


Fig. 3. Structure of a LSTM network, illustrating the flow of input, hidden states, and gating mechanisms across time steps.

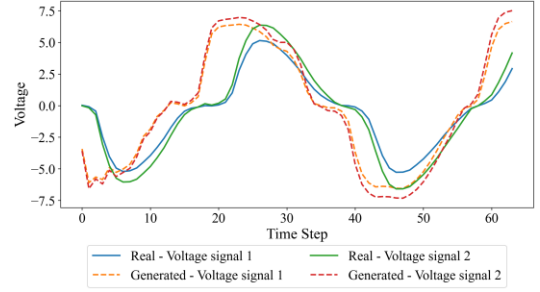


Fig. 4. A waveform overlay for real and TimeGAN-generated voltage signals for voltage signal 1 and 2.

Three well-known statistical metrics were used to evaluate the performance, including the MSE, mean absolute percentage error (MAPE), and root-mean-square error (RMSE). Their expressions are shown below:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (4)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (5)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (6)$$

where y_i is the actual value, $\hat{y}_i$ is the estimated value of critical current, and N is the number of data points.

IV. RESULTS & DISCUSSION

For our superconducting coil dataset, each measurement provided multichannel voltage and current signals under different operating conditions. We segmented these into fixed-length windows using a sliding-window approach. Normal operation and quench regimes were modelled together, resulting in one TimeGAN. As determined by tuning model and standard TimeGAN metrics, 90 sequences correspond to the optimal DS/PS configuration and provide a balanced augmentation size. This configuration preserved the temporal dynamics and statistical distributions of the experimental data. Under this configuration, the best performance is: a Predictive Score of 0.31 and a Discriminative Score of 0.15, indicating that artificial sequences were not only hard to distinguish from real sequences but also keep strong temporal consistency with real sequences [31].

Fig. 4 presents direct waveform overlays of real and TimeGAN-generated voltage signals. Synthetic waveforms

TABLE I. LSTM performance on quench prediction.

Model	Dataset	MSE	RMSE	MAPE
LSTM (no TimeGAN)	Val	0.001251	0.035373	0.038510
	Test	0.001141	0.033782	0.037011
LSTM (with TimeGAN)	Val	0.000715	0.026742	0.029839
	Test	0.001076	0.032805	0.034876

closely follow real temporal trends and quench-onset morphology, confirming high DS/PS fidelity.

The artificial datasets were then used to augment an LSTM-based predictor, configured with a hidden dimension of 96, 4 recurrent layers, batch size of 16, trained for 150 epochs with dropout regularization, and ReLU/tanh activations. The dataset was divided into 75% training, 15% validation, and 10% testing, ensuring reliable evaluation of model generalization.

Table I summarizes the performance of the LSTM quench predictor with and without augmentation. Augmentation improved validation-set performance by 23–43% and test-set accuracy by 3–6%, mainly by enhancing spike tracking near quench onset. **Fig. 5** illustrates this improvement: the baseline LSTM underestimated the minor transient and failed to capture the peak amplitude, whereas the TimeGAN-augmented LSTM reproduced the sharp voltage rise more precise, aligning closely with the measured signal. The augmented model also demonstrated earlier recognition of quench onset, providing a potential advantage for protective actions. Although the test set presents operating conditions not seen during training (different current levels, noise patterns, and sequence sources), it limits improvements. The TimeGAN-augmented model shows clearer spike tracking and improved robustness to noise. These improvements are directly relevant to quench detection and help reduce delayed identification of quench-onset transients.

Fig. 6 shows the final optimal LSTM model trained with TimeGAN-augmented data. Compared with the baseline LSTM, the augmented model significantly reduces deviations during rapid transitions and maintains consistent alignment across the entire waveform, including peaks. These results demonstrate that artificial data augmentation enhances the model's sensitivity to abrupt quench-related changes and improves its robustness to complex temporal patterns, even under the constraints of one-step-ahead forecasting.

Inference measurements were performed on an Intel Core i7-1370P CPU with 16 GB of RAM. The LSTM model runs in CPU-only mode, achieving an average latency of 0.114–0.116 ms per sequence (0.0018 ms per step, 60 steps), demonstrating fast execution suitable for real-time tracking rather than long-term forecasting. The current implementation performs only one-step-ahead prediction, meaning it improves responsiveness within the sampling interval but does not provide multi-step predictive lead-time. Future work will therefore investigate multi-step prediction horizons (5–10 steps) to evaluate whether longer foresight can be achieved without sacrificing accuracy or latency. This extension will be crucial for improving the predictive margin in fusion magnet systems where early protection is essential.

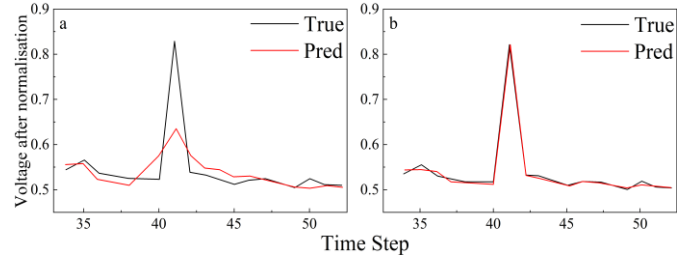


Fig. 5. LSTM prediction performance under sudden short-term changes (each time step = 0.5 ms). (a) Baseline LSTM prediction shows limited ability to capture the sharp voltage spike. (b) TimeGAN-augmented LSTM prediction more accurately follows the true signal, demonstrating improved robustness in fast transient dynamics.

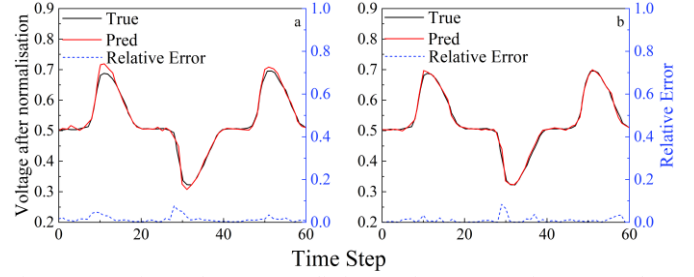


Fig. 6. Comparison of LSTM prediction performance against true voltage signals (each time step = 0.5 ms). (a) Baseline LSTM model without augmentation shows reasonable tracking but with noticeable deviations during complex waveform transitions. (b) LSTM model trained with TimeGAN-augmented data achieves closer alignment with the ground truth.

V. CONCLUSION

Accurate and Fast quench detection/prediction is crucial for protecting high-temperature superconducting (HTS) coils and enabling their application in advanced energy and transport systems such as fusion magnets and electric aircraft. However, the scarcity of experimental data limits the use of artificial intelligence (AI) and machine learning for forecasting quench precursors. This study presents a framework combining TimeGAN for artificial data generation with LSTM predictors. The results demonstrate that data augmentation significantly improves the accuracy and robustness of quench prediction. The main findings of this paper are as follows:

- TimeGAN successfully generated realistic quench sequences with strong temporal fidelity.
- Augmenting the LSTM with artificial data improved performance by up to 43% on validation sets and 6% on test sets, particularly around sharp transient events.
- The final model achieved fast inference (~ 0.115 ms per sequence), demonstrating suitability for real-time prediction.

For the superconductivity community, this paper provides a scalable approach to overcome data scarcity and strengthen AI-based protection schemes for sensitive applications. Our future work will focus on multi-step forecasting to extend predictive margins and transferability across different HTS coil geometries and operating conditions.

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